Experiment 7: Creating and Executing Your First Container Using Docker

Aim: To install Docker, create a simple container, and execute an application inside the container.

Requirements:

- Ubuntu/Linux-based system (or Windows with WSL2 enabled)
- Internet connection to pull Docker images

Procedure:

Step 1: Install Docker

Open terminal and run:

```
sudo apt update && sudo apt upgrade -y
sudo apt install apt-transport-https ca-certificates curl
software-properties-common -y
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o
/usr/share/keyrings/docker-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture)
signed-by=/usr/share/keyrings/docker-archive-keyring.gpg]
https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee
/etc/apt/sources.list.d/docker.list > /dev/null
sudo apt update
sudo apt install docker-ce docker-ce-cli containerd.io -y
docker --version
```

Step 2: Start and Enable Docker

```
sudo systemctl start docker
sudo systemctl enable docker
sudo systemctl status docker
```

Step 3: Run Your First Docker Container

1. Pull a base Ubuntu image:
2. `sudo docker pull ubuntu`
3. Run a container:
4. `sudo docker run -it ubuntu /bin/bash`
5. Inside container:
6. `echo "Hello from Docker container!" > hello.txt`
7. `cat hello.txt`
8. Exit the container:
9. `exit`

Step 4: Create a Custom Docker Image Running a Python App

1. Create a working directory:
2. `mkdir my-docker-app && cd my-docker-app`
3. Create a Python script:
4. `echo 'print("Hello from Docker!")' > app.py`
5. Create a Dockerfile:
6. `echo -e 'FROM python:3.8-slim\nCOPY app.py /app.py\nCMD ["python", "/app.py"]' > Dockerfile`
7. Build the image:
8. `sudo docker build -t my-python-app .`
9. Run the container:
10. `sudo docker run my-python-app`

Expected output:

```
Hello from Docker!
```

Result:

Successfully installed Docker, built a custom image, and executed a containerized Python application.

Viva Questions:

1. What is Docker and how is it different from virtual machines?
2. Define Dockerfile and explain its purpose.
3. What are images and containers in Docker?
4. How does Docker improve software deployment?
5. Name a few Docker commands used in this experiment.